

This chapter outlines the research design and methodological procedures employed in this thesis. The approach combines the Design Science Research (DSR) framework with qualitative interviews to develop and evaluate a data warehousing framework tailored to small and mid-sized retail businesses. The methodology is aligned with engineering sciences, where research questions often take a prescriptive or predictive form and require generalizable outcomes rather than system-specific analysis.

2.1 Design Science Research Framework

The Design Science Research (DSR) methodology was chosen for its suitability in developing practical solutions to real-world problems. In the context of this thesis, DSR provided a structured pathway for creating and evaluating an artifact—a lightweight data warehousing system—based on clearly defined needs within small retail businesses. The process was conducted in six phases:

1. **Problem Identification:** A preliminary literature review and exploratory discussions with small business owners helped identify recurring issues such as data fragmentation, lack of integration, and limited analytical capability.
2. **Define Objectives:** Based on the identified problems, key objectives were formulated: the system should be cost-effective, technically accessible, and tailored to the operational environment of SMEs.
3. **Design and Development:** A prototype was developed using SQLite and DuckDB for data storage, Go for ETL processes, and a lightweight browser-based dashboard for visualization. The architecture was chosen for its simplicity, scalability, and low overhead.

4. Demonstration: The system was tested with realistic data reflecting sales, customer activity, and product inventories, simulating a typical retail environment with multiple sales channels.
5. Evaluation: The prototype was evaluated through guided sessions with participants from the initial requirement-gathering phase. Their feedback was used to assess the tool's usefulness, usability, and alignment with everyday business needs.
6. Communication: The research process and outcomes are presented in this thesis, and the prototype source code has been made available to facilitate future development and replication.

The DSR approach ensured that the system was not merely a technical solution but a response to well-understood business challenges, allowing the results to be generalized beyond the individual participants involved in the study.

2.2 Interview Methodology

Qualitative interviews were integrated into the research design to ensure that system development and evaluation were informed by real-world use cases and grounded in the experiences of the intended users. Semi-structured interviews were selected to balance consistency across participants with the flexibility to explore unexpected insights.

2.2.1 Purpose and Rationale

Interviews were conducted in two phases. The first phase focused on understanding existing practices, data usage, and the specific challenges faced by SME retailers. The second phase, conducted after participants had tested the prototype, was designed to gather feedback on the system's relevance, ease of use, and potential improvements. This two-step approach allowed for

iterative refinement of the prototype based on real-world feedback.

2.2.2 Participant Selection

Participants were selected based on their role as decision-makers in small retail businesses, using a purposive sampling approach. Efforts were made to include businesses with varying operational models, such as physical storefronts, hybrid shops, and online retailers. This diversity helped ensure that the research findings could be generalized across different retail contexts.

2.2.3 Interview Design and Format

Each interview followed a semi-structured guide with 8 to 12 open-ended questions. The initial interviews explored topics such as data management practices, reporting needs, and technological preferences. Follow-up interviews after system deployment concentrated on system usability, perceived benefits, and areas for enhancement. Interviews were conducted in person or via video call and transcribed for analysis.

2.2.4 Analysis Method

The transcribed interviews were analyzed using thematic content analysis following Kuckartz's qualitative methodology. Responses were categorized by theme-such as usability, data needs, and feature preferences-and then condensed into key statements. These insights informed both the prototype design and the evaluation criteria, ensuring that the system addressed real operational needs.

2.3 Connecting Methods to Research Questions

The research methods were carefully chosen to address the thesis's core research questions, which reflect prescriptive and predictive dimensions of engineering inquiry:

Research Question | Methodological Basis

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What measurable benefits can small retailers gain from a lightweight data warehouse? | Evaluation interviews assessing decision-making improvements and user feedback

Which data types are most essential for improving retail business analytics and strategic planning? | Literature review and requirements interviews identifying key performance indicators

How can heterogeneous data sources be efficiently integrated using lightweight tools? | ETL implementation in Go, integrating SQLite and CSV-based sources

Which architectures and tools are best suited for cost-efficient, scalable, and user-friendly solutions? | System design decisions informed by performance testing and technical feasibility

How can findings be generalized beyond specific companies or systems? | Inclusion of diverse business types and abstraction of the solution into modular components

This methodology ensures that the findings are applicable not only to the interviewed businesses but to a broader set of SMEs facing similar data integration challenges. The study combines technical rigor with practical relevance, offering a prescriptive model for how data warehousing can be implemented effectively in small-scale retail contexts.